

Information-seeking behaviours and cognitive search strategies in different search tasks on the WWW

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Abstract

This study investigates the cognitive strategies of 80 participants as they engaged in two researcher-defined tasks and two participant-defined information-seeking tasks using the WWW. Each researcher-defined task and participant-defined task was further divided into a directed search task and a general-purpose browsing task. On the basis of retrospective verbal protocols, log-file data and observations, 12 cognitive search strategies were identified and explained. The differences in cognitive search strategy choice between researcher-defined and participant-defined tasks and between directed search and general-purpose tasks were examined using correspondence analysis. These cognitive search strategies were compared to earlier investigations of search strategies on the WWW.

Relevance to industry

Describing information-seeking behaviours and cognitive search strategies in detail provides website developers and search engine developers with valuable insights into how users seek (and find) information of value to them. Using this information, website developers might gain some knowledge as to how to best represent the content and navigational properties of websites. Search engine developers might wish to make the search and collection strategies more transparent to users. There are also design implications for the designers of web browsers.

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1. Introduction

The search behaviours and strategies of WWW users have long been of interest to WWW investigators (e.g. Catledge and Pitkow, 1995; Navarro-Prieto et al., 1999; Tauscher and Greenberg, 1997). Marchionini (1995) defined four levels of description in information seeking: moves, tactics, strategies and patterns. Moves are defined as single behavioural actions. Examples of moves when searching on the WWW might be clicking on a hypertext link, typing a query into a search engine, using the “return” button, etc. Tactics are defined as groups of behaviours. An example of a search tactic on the WWW might be typing a search query using general search terms and then using successively more specific query terms to narrow the search domain. Marchionini (1995) defined strategies as general-

ised approaches to particular information seeking problems. An example of a search strategy for the WWW might be a search engine strategy (i.e. an information-seeker who consistently uses a search engine to search for information). Finally, a pattern is defined as internalised, automatic and unconscious interactions used for all information seeking tasks, not just those on the WWW (i.e. probably something like a searching style). This particular study focuses on the specific behaviours (i.e. the moves) that constitute tactics and how these might form the basis of different cognitive strategies (i.e. strategies that also include the generalised thoughts and intentions of the users).

1.1. Information-seeking “moves” and the WWW

Despite Marchionini’s (1995) clarifications there is still much confusion in the published literature regarding

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descriptions of WWW search “moves”. Some researchers choose to describe each action (i.e. Marchionini’s (1995) “move”) whereas other researchers define groups of actions (e.g. number of links taken, number of search terms used, number of search engines used, etc.). There is a substantial body of literature that has considered each search action on the WWW (e.g. Byrne et al., 1999; Choo et al., 2000; Schacter et al., 1998; Tauscher and Greenberg, 1997). Given space constraints it is not possible to review all the literature pertaining to search moves. The studies tabulated in Table 1 have been selected as illustrations of typical search actions (as well as the of typical data collection methods used).

A number of other studies (e.g. Cockburn and Jones, 1996; Zukerman et al., 1999) have attempted to model search behaviour using mathematical modelling techniques (e.g. browsing notation, Markov modelling, longest repeating sequences, and Bayesian networks (for determining the success of search terms)). However, these sorts of techniques have largely consisted of attempts at modelling when an information seeker is going to click on another hyperlink or when they will use the browser’s navigational “buttons”. Another mathematical model, information foraging theory, based on spreading activation models also attempts to predict where (which “patches”) and how (using information “scent”) information seekers will search

for information (e.g. Pirolli and Card, 1999). Mainly using log-file data, information foraging theory approaches have attempted to model information seeking behavioural patterns on the WWW. This line of research has been used to identify aspects of the WWW (e.g. a website, a webpage, a hyperlink, etc.) as well as search behaviours such as typing/editing web addresses, using hyperlinks or navigation buttons, using search engines and query terms (e.g. Card et al., 2001).

A series of studies have also focused specifically on search engine usage (e.g. Spink et al., 2001). This line of research has been particularly useful in describing which types of search terms users of popular search engines are likely to use (e.g. Boolean search terms, single-word search terms, multiple-word search terms, whole sentences, etc.), how users change search terms (e.g. making them longer, making them shorter, changing altogether, etc.), as well as which results users choose. However, since the research is focused on search engine use it says little about how users interface with the rest of the WWW. This research is also largely based on analyses of log-files.

While this research does provide information on the range of different behaviours that WWW users adopt in their search for information, there are quite a few limitations with this data. Firstly much of the earlier work is based on a limited range of users (e.g. children, and

Table 1
Selection of studies looking at WWW search behaviours (“moves”)

Author/s	Study methods	Sample	Behaviours/“Moves”
Tauscher and Greenberg (1997)	Log-file analysis, patterns of visits and revisits to websites	<i>N</i> = 28 19 University staff and students and 9 Telecommunications company staff	First time visits; revisits; authoring of webpages; use of web-based applications; hub-and-spoke visits; guided tours; depth-first searches
Schacter et al. (1998)	Log-file analysis of search engine results	<i>N</i> = 32 School children	Analytical searching using search terms; browsing by clicking on hypertext; scan-and-select through search engine results
Byrne et al. (1999)	Concurrent verbal protocol analysis	<i>N</i> = 8 Unspecified	Use information; locate something on a webpage; provide information; configure web browser; react to environment
Hölscher and Strube (2000)	Log-file analysis and interviews	<i>N</i> = 24 Students	Following hyperlinks; using search engines; generating queries; examining search results; selecting search results; reformulating queries; going to known website directly
Choo et al. (2000)	Log-file analysis and Critical Incident interviews	<i>N</i> = 34 Unspecified	Starting (using “portals”, intranet homepages; go directly to a webpage); Chaining (following hyperlinks); Browsing (examining page headings and viewing sitemaps); Differentiating (bookmarking webpages; printing webpages; copying contents); Monitoring (revisits); Extracting (systematically searching through a website)

computer science students). Secondly, in the experimental studies, the researchers usually define the tasks that users are required to perform. Such researcher-defined tasks may specifically preclude certain types of search behaviour or strategy. Thirdly, a number of the search tasks defined in earlier studies only focus on specific elements of the WWW (e.g. search engines use, or even just searching through search engine results). Furthermore, the studies that do look at the total information-seeking behaviour on the WWW are severely limited by small sample sizes. Finally, the studies usually rely on log-file data that provides very little data on the reasons why users adopt certain behaviours and therefore it is very difficult to establish the cognitive strategy on the basis of these behaviours.

1.2. Search strategies and the WWW

Marchionini's (1995) work on electronic databases distinguished between two different categories of search strategies (of which the WWW is arguably a massive hybrid): "analytical" and "browsing" strategies. Analytical strategies are planned, systematic, specialized, goal-driven and formal whereas browsing strategies are opportunistic,

heuristic, data-driven and informal. Research into WWW search strategies also has a (relatively) long history (e.g. Catledge and Pitkow, 1995; Cothey, 2002; Fidel et al., 1999), although there are relatively few studies that have explicitly investigated information-seeking strategies.

Fidel et al. (1999) have still provided the most comprehensive published descriptions of WWW search strategies. Using verbal protocols, observational data and interviews, Fidel et al. (1999) identified seven different search strategies: "intuitive scanning" (browsing using hyperlinks, supported by a landmark webpage), "analytical" (consideration of explicit qualities of search task to arrive at explicit search query terms), "empirical" (applying rules and tactics learnt from prior searching), "known site" (going directly to a website address that is known or guessed), "similarity" (searching for information similar to what had been searched for previously), "focused searching" (focusing exclusively on the assignment topic), and "swift and flexible" (websites scanned quickly and then moving on). Fidel et al. (1999) also distinguished between two types of "intuitive scanning": "landmark searching" (using a website as a "safe" landmark for hub-and-spoke type moves) and "returning to search engine" (using the

Table 2
Studies looking at WWW search strategies

Author/s	Study methods	Sample	Search strategies
Catledge and Pitkow (1996)	3 weeks log-file analysis	<i>N</i> = 107 University staff and students	Search browsing; general-purpose browsing; serendipitous browsing; homepages as indexes
Navarro-Prieto et al. (1999)	Retrospective verbal protocol analysis	<i>N</i> = 23 Students	"Top-down" searching (searching in general area then narrowing down the search); "bottom-up" searching (searching in narrow area then broadening the area); "mixed" searching
Kim (1999)	Screen display recordings	<i>N</i> = 5 Students	"Spoke-and-hub pattern"; "breadth first" (checking multiple search results); "depth first" (going several steps away from search results before returning)
Fidel et al (1999)	Concurrent verbal protocols, observational data and interviews	<i>N</i> = 8 High school children	"Intuitive scanning" ("landmark searching" and "returning to the search engine"); "analytical"; "known site"; "empirical"; "similarity"; "focused searching" and "swift and flexible" (explained more fully in the main text)
Wang et al. (2000)	Concurrent verbal protocols	<i>N</i> = 24 Graduate students	Search engine starting; link-following; known page searching; "hub-and-spoke pattern"
Lergier and Resnick (2001)	Concurrent verbal protocol analysis of search engine results	<i>N</i> = 60 Unspecified	"Self-terminating"; "exhaustive"
Cothey (2002)	10 months log-file analysis	<i>N</i> = 206 Students	"Search querying"; "link clicking"
Ford et al. (2005)	Log-file analysis of search engine use for assigned topics	<i>N</i> = 68 Graduate students	Boolean searches; best-match searches; combined searches

search engine or search engine results as a “safe” landmark). While these search strategies are certainly more descriptive and comprehensive than other studies, one must remember that these search strategies were extracted from 8 high school children. It is difficult to ascertain the generalisability of their results to a broader WWW user population (Table 2).

The studies on search strategies have many similar limitations to those found in the behavioural “moves” studies (e.g. small sample sizes, limited sets of users, tasks defined by the researchers). In addition, previous studies have often confused a search strategy with a search task. In some studies (e.g. Navarro-Prieto et al., 1999) a search strategy is confused with a search task (e.g. a directed search task is given to participants and an “analytical” strategy emerges). Studies on search strategies are usually limited to specific experimental conditions where limitations are placed on the search context. In particular, in some experimental studies the researchers define the search tasks and thereby may rule-out certain types of search strategy. Similarly, there has been only one previous study (Navarro-Prieto et al., 1999) where participants were allowed to open multiple browser windows as part of the search strategy. This study was specifically designed to describe the search behaviours and to determine the underlying cognitive strategies determining information seeking of the WWW without these researcher-defined restrictions.

1.3. Research questions

Based on a review of relevant literature it is evident that information-seeking moves and strategies have both not been systematically investigated. In particular, previous studies have been severely limited by the size of their samples, the methods of data collection (e.g. log-file data to infer strategy) and the tasks that have been analysed (i.e. largely researcher-defined tasks). In addition, many of the research methodologies have prevented certain search moves from being used (e.g. opening new browser windows, using search moves other than search engines). The research questions for this study are thus:

- What are the dominant information-seeking strategies of WWW users in researcher- and participant-defined tasks?
- What are the dominant information-seeking strategies of WWW users in directed search and general-purpose browsing tasks?

2. Methodology

2.1. Sample

The sample consisted of 80 volunteer participants from various educational and occupational backgrounds, re-

Table 3
Summary demographic description of the sample

Occupation	N	Age	N	Highest educational level.	N
School children	7	16–20	14	High School or lower	9
Undergraduate and		21–25	47	Diploma	4
Graduate student	46	26–30	18	1st degree	39
Academic staff	7	31–35	1	Higher degree	26
IT Professional	7				
Other Professional	3	Gender	N	Home language	
Semi-Professional	10	Male	50	English	51
		Female	30	Non-English	29

cruited using non-probability snowball sampling (Fife-Schaw, 1997). Snowballing involves approaching participants known to the researcher. These participants were then requested to suggest anyone suitable that they knew. Participants were undergraduate or postgraduate students ($N = 46$), academic staff ($N = 7$), school children ($N = 7$), IT professionals ($N = 7$), or other professionals and semi-professionals ($N = 13$). Fifty of the participants were male, 51 of the participants spoke English as a first language, and the majority of participants ($N = 65$) had at least a bachelor's degree. The demographic summary of the sample is given in Table 3.

Participants had a range of experience with computers and the WWW. Participants had been using a computer for an average of 97.81 months ($SD = 52.96$) and spent an average of 20.44 ($SD = 15.81$) hours per week using a computer. While all the participants had used the WWW at least once there was quite a wide range of experience. On average, participants had been using the WWW for 36.86 months ($SD = 20.22$) and spent 7.35 h ($SD = 7.86$) per week using the WWW.

2.2. Procedure

Following a structured interview where participants were asked a series of biographical questions as well as questions on their prior use of computers and the WWW, the participants had to complete four tasks. Using a one-group repeated measures design, the ordering of the four tasks was randomly assigned to control for any learning effects. Following Marchionini and Schneiderman's (1988) task definitions the four tasks consisted of two directed fact-finding tasks and two general-purpose browsing tasks. In each of these task conditions, one task each was defined by the researcher (i.e. “Find Bill Clinton's mother's maiden name” for the directed search task and “Find all information on the relationship between carbon monoxide and desertification” for the general purpose browsing task) and one task each was chosen by the participant. Participants were able to select their preferred browser (only Internet Explorer 5.0 or Netscape Communicator 4.5 were chosen as these were the latest versions at the time of data collection), but each participant started each search

task from the same, researcher-defined, homepage. The participants were given a maximum of 15 min each for each task and could terminate the search at any point before the 15 min were up if they felt they had completed the task or they wished to stop. Each task was recorded using Lotus ScreenCam. The ScreenCam recordings were then played back where retrospective verbal protocols were collected after each task to determine participants' reasons for making behavioural moves. Client-side log-files and observational data were also recorded. Similar methodological approaches have been used in other research to uncover WWW search strategies (e.g. Fidel et al., 1999; Navarro-Prieto et al., 1999). The data was collected between November 1999 and December 2000.

2.3. Analyses

The search moves were obtained by examining a combination of data sources: client-side log-files, observational data, ScreenCam recordings and the retrospective protocols. The client-side log-files provided the time for each behavioural event and therefore a chronological recording of sequences of events. From these sequences of events a coding scheme was developed that described the search moves. The search moves were then organised into chronological sequences and compared to Marchionini's (1995) theoretical model of the search process. From an inspection of clusters of moves, regular sequences of behavioural moves (i.e. tactics) were identified. After three iterations 78 tactics emerged.

Next, cognitive strategies were identified. Cognitive strategies were derived on the basis of combining information from the search tactics with information on the motivations, intentions, and rationalisations behind each search process that were extracted from the retrospective verbal protocol data. An inspection of participants' intentions at each decision-making point provided the basis for determining whether similar tactics and moves had the same intentional root. Where a participant expressly indicated that they were changing their strategy, these portions of the chronological order were analysed separately. Once each portion of the chronological behaviour had been assigned an intention or motivation, then these were compared for similarity of behaviour and similarity of intention. Congruency between intentions and behaviour was a strong indicator of a cognitive strategy. If there was incongruence between behaviour and intention, these portions were carefully examined to see if the sequence could be broken down further or whether there were different behaviours to achieve the same intention. The cognitive strategies were assigned for each of the four tasks. Following Fidel et al. (1999) it was possible for participants to use more than one strategy during a search task. Statistical comparisons were made across the four task conditions using χ^2 tests and correspondence analysis. Correspondence analysis is a variant of principal components analysis that provides a low-dimensional graphical

representation of the rows and columns of a contingency table (Greenacre and Hastie, 1987). In correspondence analysis the dimensions have no meaning unlike factor analysis or multidimensional scaling. Rather, correspondence analysis provides a means of graphically representing the angles between the data points.

3. Results

3.1. Search moves and tactics

There were 28 types of search moves identified after several iterations from an analysis of the log-file data and retrospective verbal protocols. Following Marchionini's (1995) general search model the search moves were categorised into five broad areas: task initiation behaviours, search terms, sustaining search behaviours, task termination behaviours and unusual behaviours. These search moves are given in Table 4. From Table 4 it is evident that these search moves are not described at the typical level of abstraction (i.e. "click on hyperlink", "type in query"), but at a level of abstraction that implies some level of intentionality.

These search moves were then organised into chronological sequences that constituted 78 different tactics. Space does not allow for the presentation of all 78 tactics but these are summarised in Fig. 1. The search process is represented in four steps: initiation, search terms (formulating query and examining results), sustaining and terminating.

3.2. Cognitive search strategies

Twelve different cognitive search strategies that reflected both the retrospective verbal protocol data and groups of search behaviour patterns were identified. It was evident that participants' frequently changed their search strategy during the course of a single task. In fact, some participants changed their search strategy as many as three times for one task. In some cases these changes were incorporated as part of a larger strategy (e.g. "Sequential player" strategy or the "Parallel player" strategy) but there were a number of times when the participants were genuinely changing their search strategies. The change of strategy would have happened if participants felt either that a particular search strategy was unsuccessful, or they wanted to use different search strategies simultaneously in different browser windows, or because they were extending or refining existing search strategies. The cognitive search strategies are described below.

3.2.1. "Safe player" strategies

The "safe player" strategies were characterised by four different types of strategies. Each of these strategies involved the participant using a cognitive strategy that enabled flexibility around a set of parameters familiar to the participant. The participant attempted to produce

Table 4
Search moves across the four tasks

Task initiation behaviours

Going directly to a known/preferred search engine
Using browser “search” button to “select” search engine
Type in website address/guess website address
Use history list, favourites menu, or bookmark to find previously visited webpage
Follow links from homepage

Search terms (only if a search engine is selected)

Search terms very specific to the task
Search terms general to the search task
Boolean logic operators to limit/broaden search parameters
Use search engine categories to narrow search domain
Use broad search terms not emergent from the search task

Sustaining search behaviours

Hub-and-spoke pattern from a landmark webpage
Guided tour (website defined “tour”)
Use “home” or “search” button to return to “start” of the search
Use scroll-bar or history list to reposition within the search task
Open new browser window to conduct same/similar search in parallel
Open new browser window to conduct different search in parallel
Change search engine for the same/similar search
Going through search engine results pages

Terminating behaviours

Follow short path (fewer than six links) to an answer/termination point
Follow long path (more than five links) to an answer/termination point
Find answer/termination point from a hub-and-spoke landmark webpage
Find answer/termination point from a guided tour
Giving up/resigning oneself to no answer being found

Unusual search behaviours

Physically altering aspects of web address by removing/adding extensions
Using the “find” dialogue to search within a webpage
Following a new browser window that opens automatically and not realizing this
Makes a physical record of webpage (e.g. saved to bookmarks, saved to word processor)
Spelling problems

search circumstances that were familiar and had (possibly) worked in previous web searches. The participant seldom strayed from the known WWW domain or known search terms that defined the task. The intention in the “Safe Player” strategy was for the participant to remain within a safe or familiar search environment, and not to wander too far into “unknown” search territory.

3.2.1.1. Broad first. In this strategy, the participant went to a search engine that was known to them. If they were able to recall the web address of a specific, favoured search engine, then they would type in this address in the address command line. If not, they either clicked on the search icon on the task homepage to select their favoured search engine, or clicked on the browser search icon to let the web browser take them to the web browser selected search engine. In their favoured search engine they would use one or more general search terms defined by the search task. This strategy is regarded as broad since the search terms were directly relevant to the task but broad enough to

obtain a wide range of search results. Depending on the exact search terms and the particular, favoured search engine, this strategy would yield a myriad of different results. In some cases, the search engine was able to return a result where the participant could follow a single, short or long path to the answer. In other cases, the participant may have had to follow a hub-and-spoke pattern before finding an answer. In some cases, the participant may have had to go back to the search engine to redefine the search terms. In this particular strategy, they would have started with general terms and then used more detailed, task-specific terms. In certain cases, participants used Boolean logic terms to narrow the search domain. The intention behind the Broad first strategy was to remain within a suitably familiar territory (i.e. to use terms broad enough to be familiar and yet within the domain of the search task).

3.2.1.2. Search engine narrowing down. In this strategy, the participant selected a search engine on the basis of narrowing down the search domain. They chose a search engine based on the known (or suspected) attributes of the search engine. They were essentially using the search engine to narrow the field of search. Alternatively, they accessed their favoured search engine in one of the manners described in the “Broad first” strategy. However, instead of choosing one or two broad terms (as defined by the task), they chose a search engine category to narrow the search. The category could have been a general category, or the advanced search option. In both of these choices they had the same intention, to narrow the field of search before using search terms. The strategy was “safe” in that the participant did not have to risk going into unfamiliar territory, but rather used browser categories that appeared to logically narrow the task search domain. This narrowing down strategy may have had a number of outcomes. The narrowing down may have led to them finding an answer before search terms were used. This would certainly be the intention of most search engine website designers. More likely, however, was that they would have had to use search terms to find an answer. If they believed that the search engine had been successful in narrowing the field of search then they would have used task-specific search terms. If they believed that the search engine was unsuccessful in narrowing the search domain, they would have used broad search terms in a similar manner as described in “Broad first” strategy. At this point, the search process followed a similar pattern to the “Broad first” strategy.

3.2.1.3. “Search engine player” strategy. In the “search engine player” search strategy, participants used different search engines to conduct the same search. They used this strategy to search different portions of the WWW or to search the WWW using a different search algorithm (provided by the search engine) with the same search terms. They mentioned that different search engines have different capabilities. For example, one participant claimed that NorthernLight (a particular search engine) was good

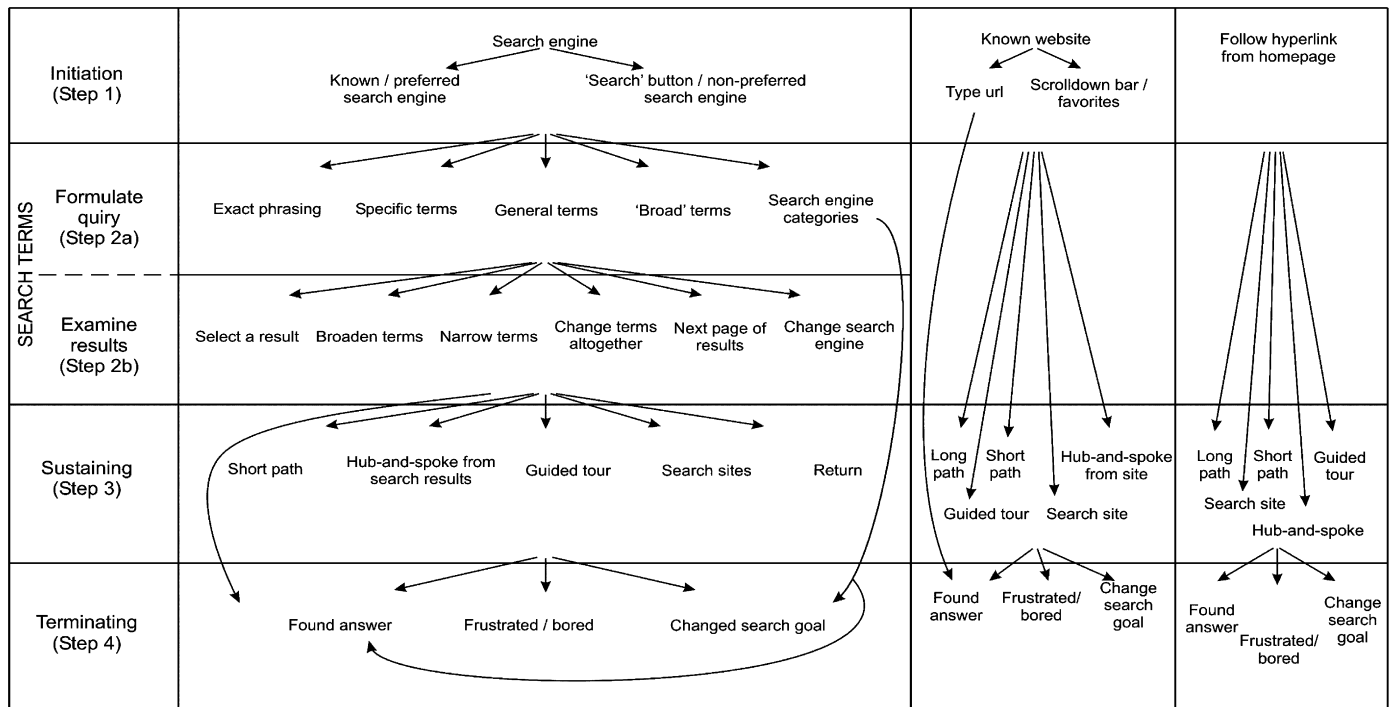


Fig. 1. WWW information-seeking process and tactics.

for academic and research searches, while Yahoo (a different search engine) concentrated on American webpages. They would use different search engines depending on whether the capabilities of a search engine matched the particular search task. They would then switch to a different search engine if the initial search engine was unsuccessful. A variation in this strategy would be when a participant used a search engine that returned search results from multiple search engines (meta-search engines). In effect they were using different search engines to conduct the same search, but the changing of the search engine was automatic. This variation had the advantage of direct comparison between the different search engines. They were able to identify common and unique search results for multiple search engines. These variations were regarded as “safe” since they were remaining with the “known” (or inferred) capabilities or bounds of the search engines.

3.2.1.4. Known address search domain. In this strategy, the participants went directly to a webpage that was not a search engine, but that could be used as another suitable portal for the particular search task. If they were able to recall the webpage address then they would have typed in this address in the web browser address command line. They also stated that they accessed the known webpage through their bookmark/favourites menu or by using the scroll bar if they had been to the webpage recently. Alternatively, if they did not know the web address of the particular target webpage, they may first have used a search engine to get to the webpage. The “Known address search domain” was not the webpage where they expected

to find the answer, but rather the point from which they intended to embark on their search task. The webpage had been selected because it defined a known search domain or contained links to webpages that may have had the answer. They usually followed a hub-and-spoke search pattern from the known webpage (using the known webpage as the landmark). In some cases, they may have been able to go directly to the answer (either through a single, short path or a single, long path) without having to engage in hit-and-miss behaviour.

3.2.2. “Parallel player” strategy

In the “parallel player” strategy, the participant deliberately opened multiple browser windows to conduct different searches simultaneously. At least one new browser window was opened early in the search task session. They usually attempted a different search strategy in each browser window. Typical combinations were one or more of the following:

- (i) The participant used the same search engine but different search terms in each browser window. Usually they searched in one browser window with specific search terms and in another browser window with general search terms. If they were unclear on the meaning of the terms, it was possible that they used general search terms in more than one browser window to get clarity on the meaning of the different search terms.
- (ii) The participant used the same search terms but with different search engines. As mentioned previously in the “search engine player” strategy, participants

argued that different search engines search different portions of the WWW (or collected search results using different search algorithms).

- (iii) The participant used a browser window and general search terms to conduct a “broad first” strategy and another browser window to conduct a “known address search domain” strategy.
- (iv) The participant used a browser window to conduct a search engine search and another browser window to go straight to a known webpage address.

If a search engine search was conducted then they were likely to follow a hub-and-spoke pattern from one or more of the browser windows. They had to constantly switch between browser windows to check on the progress, as a reminder of the content or to continue with some other sustaining search behaviour. In certain cases, they were distracted away from other browser windows that were open if one browser window appeared more promising. The intention behind the “parallel player” strategy was to use the capabilities of browsers to conduct multiple types of search tactics concurrently.

3.2.3. *Link-dependant strategy*

In the “link-dependant” strategy, the participant relied solely on hyperlinks from the homepage to get from one webpage to another. Typically, this strategy would start from the task homepage and then clicking on one of the hyperlinks in the homepage. In extreme cases of this strategy participants were stuck without a hyperlink and had to use the “back” button on the web browser to get to a webpage that had hyperlinks. In these extreme cases, this may possibly be the default strategy for the participant who knows very little about search engines and WWW searching. However, even participants who were familiar with search engines could sometimes use this strategy if they believed that hyperlinks from the homepage could assist in finding the search answer. The intention behind the “link-dependant” strategy was one of two types. In the first type, the participant used links from the homepage in searching for information on the particular task. In the second type, the participant used links from the homepage possibly because they were unfamiliar with search engines and felt more comfortable using links than launching into the unknown. It must be stressed that this strategy was not the same as link-dominant behaviour. Link-dominant behaviour describes the dominance of using a hyperlink to get from one webpage to another in normal WWW searching behaviour. A participant using a “link-dependant” strategy would only able to use hyperlinks and cannot conceptualise making an illogical leap to a webpage that is not hyperlinked.

3.2.4. *“To-the-point” strategy*

In the “to-the-point” strategy, the participant went to their favoured search engine (in one of the ways described in the “broad first” strategy). This strategy differed from

any of the “safe player” strategies in that they first used quite specific search terms. They were not attempting any “safe” narrowing strategy (such as search engine categories or broad terms) but rather, their intention was to get directly to the answer straight away. They used the exact words or a phrase that encapsulated the search task. In certain instances, they used Boolean logic terms to narrow the search domain or to exclude certain search domains. Depending on the exact search terms and the particular, favoured search engine, this strategy (like the “safe player” strategies) yielded a range of different results. Again, it was possible for the search engine to return a result where they followed a single, short or long path to the answer. In other instances, they may have had to follow a hub-and-spoke pattern before finding an answer. If the hub-and-spoke pattern proved unsuccessful, they may have had to go back to the search engine to redefine the search terms. In this particular strategy, they started with specific search terms and so used general terms to broaden the search domain. In many ways, the “to-the-point” strategy looked similar to a “safe player” strategy at this point. This strategy was characterised however, by the search terms gradually becoming broader and broader as the specific search terms proved unsuccessful. The intention behind the “to-the-point” strategy was to find an answer as quickly as possible by being specific, rather than general.

3.2.5. *Known address strategy*

In the “known address” strategy, the participant went straight to a known site that was not specifically a search engine. Where they recalled the webpage address of the known site, they entered the web address in the web browser address command line. Where they had forgotten the webpage address, then they may first have to go to a favoured search engine (in a manner described in the “broad first” strategy) and search for the target webpage address. Some participants also mentioned that this webpage could be found in the bookmark menu of their own computer. This search strategy differed from the “known address search domain” strategy in that participants using this search strategy knew exactly where in a known website the information that they are looking for was to be found. In the “known address search domain” strategy, they still had to search for the target information once they had reached the known webpage. In the “known address” strategy, they had usually been to the website before and knew exactly where the information could be found.

3.2.6. *“Sequential player” strategy*

The “sequential player” strategy is a combination of a “safe player” and “to-the-point” strategies. In the “sequential player” strategy the participant made a rapid change from a “to-the-point” strategy to a “safe player” strategy, or vice versa. In a “safe player” strategy there was a gradual broadening of the search terms. In the “sequential player” strategy the participant went straight

from general search terms to exact words or a phrase without the intervening specification of terms. Alternatively, they first tried exact words or a phrase and then attempted very general search terms. Alternative combinations were possible with the “known address search domain”, the “search engine narrowing down” and the “known address” strategies. Participants who were trying to confirm specific information also used this strategy. In the general search they may have found some indication of an answer and so used a specific search to verify this answer.

3.2.7. *Deductive reasoning strategy*

In the “deductive reasoning” strategy, the participant usually started in a similar manner to that described in the “broad first” strategy to get to a search engine. However, in this search strategy the search terms are quite different. They attempted to use search terms that appeared unrelated to the search task. These usually appeared to be illogical search terms, but a closer inspection of the reasoning behind such search term selection usually reveals that they were taking an analytical approach to the search task. The search terms appear illogical because they had thought beyond the terms supplied at the start of the task. The “deductive reasoning” strategy was used when they felt that the search terms were either too broad or too specific. In this strategy, they attempted to reason through the task, based on their own conceptualisation of the problem domain (or sometimes based on what they interpret from engaging in the search task).

3.2.8. *Secondary search strategies*

In addition to the ten cognitive search strategies describe above there were two secondary search strategies. Secondary search strategies are those strategies that did not have a fixed initiation step.

3.2.8.1. “Virtual tourist” strategy. In the “virtual tourist” secondary strategy, the participant followed a predefined path through a particular website in pursuit of the answer. This strategy could have followed any of the “safe player”, “parallel player”, “to-the-point” “sequential player”, “link-dependant”, “deductive reasoning” or “known address” strategies. In the “virtual tourist” strategy, participants followed a predefined path rather than choosing a particular hyperlink that would take them straight to the answer. In this respect, the cognitive reasoning behind this search strategy was quite different to a hub-and-spoke or a single path pattern. In the “virtual Tourist” strategy they move through some predetermined (by the website designers), linear path through the website. The return to the familiar, landmark page is often beyond their control.

3.2.8.2. Parallel hub-and-spoke strategy. In the “parallel hub-and-spoke” secondary strategy the participant opened up a hyperlink in a new browser window. In the “parallel player” strategy, the opening of a new browser window was

to launch a new search whereas in the “parallel hub-and-spoke” strategy the intention was to open up a particular hyperlink as a new webpage in a hub-and-spoke type pattern. The difference between a “parallel hub-and-spoke” strategy and other hub-and-spoke patterns was that they may have wanted to follow a path in one of these new browser windows and then return to the landmark webpage (the webpage from where the new browser window was opened) while still keeping the end result of each “spoke” open. Keeping the end result of each “spoke” open has the advantage of effectively saving the search result for perusal at a later stage. These new browser windows could have been reviewed at any point in the search task without having to repeat a search path.

3.3. *Cognitive search strategies across the four tasks*

The frequency of strategies across the four tasks is presented in Table 5. A χ^2 test of association on this contingency table shows that the strategies were unevenly distributed across the four tasks and that this difference is statistically significant ($\chi^2 = 138.86$ (d.f. = 33,1), $p < .01$). The “known address” strategy and the “virtual tourist” strategy were only found in the two participant’s choice tasks. The “broad first” strategy was common to all four tasks but was found more frequently in the two directed search tasks. The “search engine narrowing down”, the “search engine player”, the “parallel player” and the “to-the-point” strategies were also common to all four tasks, but were found more frequently in the two researcher’s choice tasks. Conversely, the “known address search domain”, while common to all tasks, was found more frequently in the two participant’s choice tasks. However, the χ^2 statistic does not indicate where there are significant differences, only that there are statistically significant differences somewhere within the contingency table.

In order to graphically examine the relationship between the four tasks and the cognitive search strategies more closely, correspondence analysis was used. One dimension explained 70% of the inertia and two dimensions explained 89% of the inertia. In the two-dimensional description the correspondence analysis in Fig. 2 demonstrates which strategies are associated with which tasks. The researcher’s choice, directed search task was most closely associated with three strategies, the “broad first”, the “search engine narrowing down” and the “deductive reasoning” strategies. The researcher’s choice, general purpose browsing task was most closely associated with five of the cognitive search strategies. These are the “search engine player”, the “sequential player”, the “to-the-point”, the “parallel player” and the “parallel hub-and-spoke” strategies. It is also clear that these two tasks were more closely associated with one another than with the two participant’s choice tasks. The participant’s choice, directed search task was most closely associated with two strategies, the “known address search domain” and the “known address” strategies. It is clear that the participant’s chose to search in

Table 5
The frequency of the twelve strategies across the four task conditions

Strategy	Directed Search researcher's choice (N)	General purpose researcher's choice (N)	Directed search participant's choice (N)	General purpose participant's choice (N)
Safe player "broad first"	35	20	35	29
Safe player "search engine narrowing down"	18	19	14	13
Safe player "search engine player"	6	9	1	4
Safe player "known address search domain"	1	2	26	18
"Parallel player"	7	12	1	5
"Link-dependent"	1	3	2	4
"To-the-point" searcher	9	17	5	5
"Known address"	0	0	9	4
"Sequential player"	6	13	5	6
"Deductive reasoning"	3	6	6	1
"Virtual tourist"	0	0	6	18
"Parallel hub-and-spoke"	2	11	4	6
Total N strategies	88	112	114	113

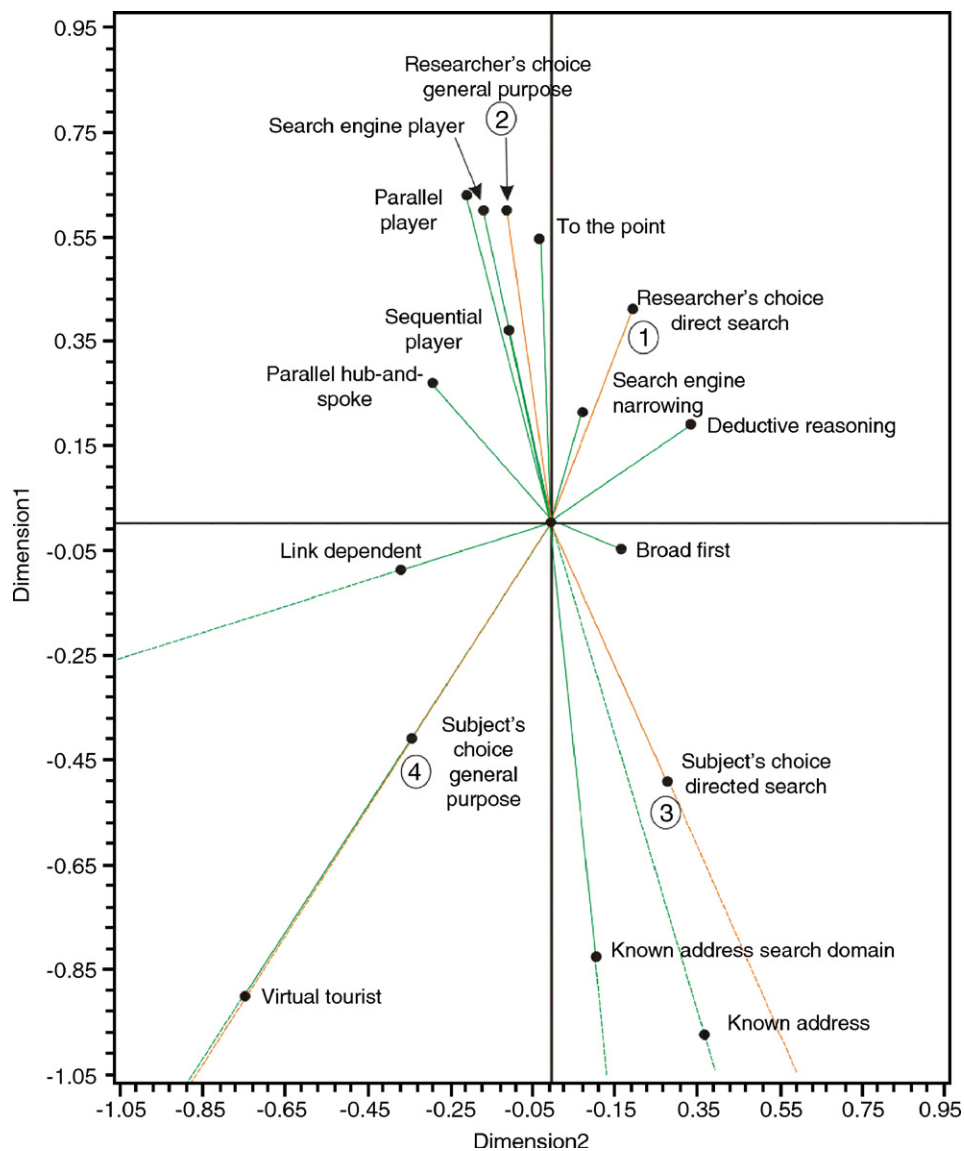


Fig. 2. Correspondence analysis of the cognitive strategies across the four tasks.

areas where they had most familiarity (either domain knowledge or knowledge of a specific site). The participant's choice, general purpose browsing task was also most closely associated with two strategies, the “link-dependant” and the “virtual tourist” strategies.

4. Discussion

4.1. Comparisons with earlier studies on search strategies

Marchionini's (1995) two search strategy categories, analytical and browsing, were defined by their level of goal-directedness, formality, and planning. The implication in these two categories of searching is that the categories of strategies refer to different types of tasks. A number of studies have used this broad categorisation to define search strategies (e.g. Cothey, 2002), while other researchers have defined strategies according to the goal-directedness of the task (e.g. Catledge and Pitkow, 1996). Bilal (2000) defined the analytical search strategy as using search terms and the browsing strategy as using search engine categories. While these search strategies were at least inferred from an analysis of the search behaviour, the “strategies” referred to only one part of the search process (e.g. using a search engine). In a similar manner to Lergier and Resnick (2001) it is therefore more likely that these “strategies” referred to search tactics. Similarly, Lazonder (2000) distinguished between a “search engine” strategy and a “web address” strategy. It is possible to distinguish between these two categories of strategy in this current study. The “web address” category would be similar to the “known address” and the “known address search domain”. However, the “search engine” strategy would describe a number of different strategies found in this current study. As was observed in this study and was reflected in the definition of the search strategies, there was often a fluid transition between search engine use (the implied derivation of many analytical strategies) and browsing through websites, webpages and search engine results (the implied derivation of browsing strategies). In this study, the distinction between analytical and browsing strategies seems superfluous. Individuals did not appear to change strategies once search engine results are returned. Rather, the way in which they looked at search engine results was constrained by the overall strategy that they had in mind about the best way to conduct the search. The transition from query formulation to results examination was seamless for the WWW searcher. Unlike Catledge and Pitkow (1996), no participants demonstrated that they had gone to the trouble of developing their own webpage with hyperlinks to webpages of interest, although this “strategy” is still technically possible. This may have been an artefact of the research design or the sample.

The conceptualisation of cognitive search strategies from Fidel et al. (1999) comes closest to how search strategies were defined in this current study. The “known address”

strategy in this current study was most similar to the “known site” strategy from Fidel et al. (1999), although they are defined slightly differently. In this study, participants could use a search engine to get to the known website if they could not remember the web address. In Fidel et al.'s (1999) study, their definition was not so obvious. A known website was one where their participants could guess at a web address on the basis of their knowledge of the structure of web addresses (not just websites that were known to the school children). The analytical strategy was essentially defined in the same way as the “deductive reasoning” strategy in this study, although this study provided more clarity to the definition by providing some of the cognitive reasoning for the strategy. The similarity strategy was not defined as such in this study since it was not possible from the retrospective protocol data to distinguish the “known address” strategy from the similarity strategy. Some participants stated that they had done something similar before and were attempting to repeat this information-seeking process. In essence they were repeating some other search strategy and so it was this search strategy that was captured rather than a “similarity” strategy. The browsing strategy was also not defined as a separate strategy in this study. The closest that this study gets to a browsing strategy would be the “link-dependant” strategy. Browsing was defined as one of the behaviours that participants used to traverse from one webpage to another. It was not defined as a strategy in itself.

Fidel et al. (1999) distinguished between two general searching strategies and two search strategies to allow participants to move to “safe” and familiar portions of the WWW. It is interesting to note that Fidel et al. (1999) also realised that participants were systematically choosing to remain within “safe” and familiar areas. The two general search strategies were “focused searching” and “swift-and-flexible” searching. However, even in these strategies, Fidel et al. (1999) described a number of ways in which these were carried out. For example, in the “focused searching” category, Fidel et al. (1999) noted that one participant went directly to a web address (by typing the school's URL) whilst the majority of the other participants used search engines. In addition, Fidel et al. (1999) appeared to be confused about what they meant by “focused searching”. For example, they noted that participants changed the specific search topic or recorded information only broadly applicable to the topic. These two general strategies were too indistinct to be described in this current study. The two strategies that allowed participants to move to familiar portions of the WWW were “landmark searching” and “returning to a search engine or starting a new search”. These “strategies” were search behaviours observed within the “safe player” strategies described in this current study. They were not described separately as different strategies, but rather were incorporated within cognitive search strategies that lasted for a whole task (or for a large portion of the task).

Navarro-Prieto et al.'s (1999) strategies also bear some similarity to the strategies defined in this study. The top-down strategy is very similar to the “broad first” strategy, “search engine player” and “search engine narrowing down” strategies found in this study. Navarro-Prieto et al. (1999) defined the top-down strategy as searching within a general area first and then gradually narrowing down the search when the broad search was unsuccessful. Similarly, the bottom-up strategy was most similar to the “to-the-point” strategy found in this current study. Navarro-Prieto et al. (1999) defined the bottom-up strategy as users searching using specific keywords provided with the task instructions. The mixed strategy was defined as two different strategies in this study. If participants utilised top-down and bottom-up strategies in different web browser windows, this would have been defined as one example of the “parallel player” strategy. If participants used these two strategies in the same browser window, this would have been defined as a “sequential player” strategy. Likewise, Kim's (1999) “broad first” strategy is similar to the search strategy of the same name from this current study whereas Kim's “depth first” strategy bears some similarity to the “to-the-point” search strategy from this current study and the “hub-and-spoke” pattern is found in a number of “safe player” search strategies in this current study.

It is also evident that through research design some researchers have deliberately excluded certain types of search strategies. For example, Card et al. (2001) deliberately prevented any of the “parallel player” or “parallel hub-and-spoke” strategies by not allowing their participants to open new browser windows. Yet, it was perfectly obvious in this current study that this was an effective search strategy. The cognitive search strategies, as they were defined in this study, refer to more than behavioural descriptions of the task. Rather, through a careful analysis of the retrospective protocols, the search strategies captured something of the intentions and rationale of the participants. It should be clear that participants could use the same behaviour but have a different intention, and therefore a different strategy in mind. It is this combination of the behavioural manifestation and the cognitive reasoning of the participant that provides structure and coherence to the search strategy definitions described in this article.

4.2. *Type of task and cognitive search strategies*

It is quite clear that the methodology of previous studies of researcher-defined tasks systematically excluded “known webpage” behaviours and strategies. Even in Fidel et al.'s (1999) study, the teachers and/or researchers provided the assignments and web addresses. In this study, attempts without knowledge were recorded as (best, educated) guesses rather than “known webpages”, since the participants did not actually know the content of the webpages but were attempting to infer the webpage address from the task description. The analyses of the cognitive search

strategies across the four task conditions also showed a number of interesting trends. The most obvious finding was that the four “safe player” strategies (except for the “search engine player” strategy) were the dominant strategies in each of the task conditions, particularly the “broad first” strategy. Based on log file analyses Tauscher and Greenberg (1997) have suggested that users return to webpages that are familiar. In addition, Fidel et al. (1999) identified two tactics that would return users to familiar areas, “landmark searching” and “returning to a search engine or starting a new search”. Participants who were classified as using one of the “safe player” strategies mentioned that they were using certain webpages as “landmarks” and then following hub-and-spoke patterns from these “landmarks” in order not to get lost. Subsequently, they rarely moved more than few steps away from these “landmark” webpages. The “broad first” strategy enabled the participants to define the broad parameters of the search rather than the “risky” behaviour of being too specific. The “search engine narrowing down” strategy enabled participants to use the perceived attributes of the search engine to narrow the search. This involved either choosing search engines because of their perceived usefulness for the particular task, or using the search engine's categories to narrow the search parameters.

No previous studies have systematically investigated information strategy differences between researcher-defined and participant-defined information-seeking tasks, so it is not possible to make direct comparisons (although Kim (1999) looked at differences in search activities; e.g. average number of sites visited and average number of search tools used). From Fig. 2 it is evident that the two participant-defined tasks are associated with distinct search strategies when compared to the two researcher-defined tasks. The researcher-defined tasks are associated with the more analytical (i.e. “deductive reasoning”, “search engine player”, “search engine narrowing down”, “sequential player”, “parallel player” and “parallel hub-and-spoke”) and “safer” strategies than when compared to the more intuitive and personal (i.e. “known address”, “known address search domain”, “link-dependant” and “virtual tourist”) strategies used for the participant-defined tasks. In researcher-defined tasks it was evident that participants used strategies that would take them the most direct path to an answer (i.e. the “to-the-point” strategy) or that relied on the use of search engines. This implies that studies that rely on researcher-defined tasks to study search strategies (e.g. Fidel et al., 1999; Navarro-Prieto et al., 1999) may produce misleading results in terms of the full range search strategy possibilities.

Comparisons with previous studies that have investigated differences between directed searching and general-purpose searching showed some interesting links and differences. Cove and Walsh (1988) implied that general-purpose tasks would encourage browsing strategies. However, they also implied that directed search tasks could also be undertaken through a “search browsing” strategy.

In this study it was obvious that the strategies could not simply be divided into browsing and analytical strategies. It was often the case that participants moved seamlessly from “analytical” search engine use to “browsing” using hub-and-spoke behaviours within the same search strategy. In two researcher-defined tasks Schacter et al. (1998) noted that searching on the WWW was dominated by the use of search engines (a similar finding to this current study) and that analytical behaviour was found most frequently in the directed search task. Browsing behaviour was found most frequently in the general-purpose task. Choo et al. (2000) found more instances of revisiting known sites and using search engines to extract information in the directed search task. This would be similar to the results of this study where participants used search engines if they were not sure where to start (particularly in the researcher’s choice task), and revisited known websites, especially in the participant’s choice task. In the general purpose browsing task (referred to as an “informal search” and “conditioned viewing” in their study) Choo et al. (2000) found more instances of search engine use to extract information, revisiting known websites, browsing sitemaps and browsing webpages. Since Choo et al. (2000) were investigating tasks that participants had chosen it is not surprising how similar their behaviours were to the strategies described in this study. While the previous results are an obvious over-simplification, in this study the “broad first”, “deductive reasoning” and “search engine narrowing down” strategies were most closely associated with the directed search task and the “link-dependant” and “virtual tourist” strategies were most closely associated with the general purpose tasks.

4.3. Limitations and conclusions

The most obvious limitation was that this study was conducted in a laboratory setting. This type of setting may have prevented participants from attempting certain types of information-seeking strategies (e.g. using bookmarks and history lists, unusual homepage designs, or homepage “portals”) and the presence of the researcher may have discouraged participants from searching for certain types of familiar information (e.g. illegal websites, pornography, etc.). The time constraints may also have limited the search strategies available to participants. It was also evident that participants sometimes changed the purpose of the search (e.g. a general-purpose task sometimes became a directed search task once specific general information had been found). One must also remember that these results apply specifically to information seeking situations where the information being sought was not known a priori, and not necessarily to general WWW use that may not be related to seeking for information (e.g. news updates, webpage authoring, etc.).

It is also possible that other individual variables might have had a stronger influence on the choice of cognitive search strategy. In particular, there is the possibility (based on previous studies) that prior experience or expertise with

the WWW (e.g. Hargittai, 2002; Hölscher and Strube, 2000; Lazonder, 2000) and subject-area knowledge (e.g. Hölscher and Strube, 2000) would have a marked effect on the search strategy adopted by a participant. This data has been analysed and is intended for future publication. Similarly, other individual factors such as cognitive style and general problem-solving style might also influence the choice of search strategy (Ford et al., 2005; Kim, 1999; Kim and Allen, 2002).

No value judgements have been placed on any of the search strategies in terms of their effectiveness. In fact, in the participant-choice tasks it is extremely difficult to independently determine effectiveness or success. Since these tasks were the choice of the participant, it was not possible to control for task difficulty. In the participant-choice tasks the participants decided when the tasks were complete so effectiveness measures such as correct answers, length of time and number of webpages visited does not necessarily bear any relationship to effectiveness. As a consequence, it is not possible to be prescriptive about which search strategies are most effective.

Despite these limitations, the results suggest that the choice of search strategy was dependant not only on type of search task that they were attempting but also on whether the task was defined by the researcher or the participant. The results imply that aspects such as subject-area knowledge and past experience with the WWW play a role in determining the information seeking strategies that one would use. In particular, if participants defined the tasks they were likely to use “known address” or “known address search domain” strategies for the directed search task and “link-dependant” or “virtual tourist” strategies for the general-purpose browsing task. The results also suggest that at least two of the cognitive search strategies (i.e. “search engine narrowing down” and “search engine player” strategies) were utilised by participants to exploit the different perceived qualities of search engines. In particular, participants chose search engines on the basis of qualities such as the search algorithm, the search terms or phrases that could be used, how the search engine collects webpage listings, and how the search results are ranked. In a separate paper based on the same sample (Thatcher and Greyling, 2003) it was found that these perceived qualities were often misconceptions of the true capabilities of the search engines. Participants may have been utilising search strategies that had no validity based on the true capabilities of the search engine. Certainly, search engine interface designers could do more to make the search capabilities of the search engine clearer to users.

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